

Cellular Respiration:

Obtaining Energy from Food

Unit 1; Chapter 6

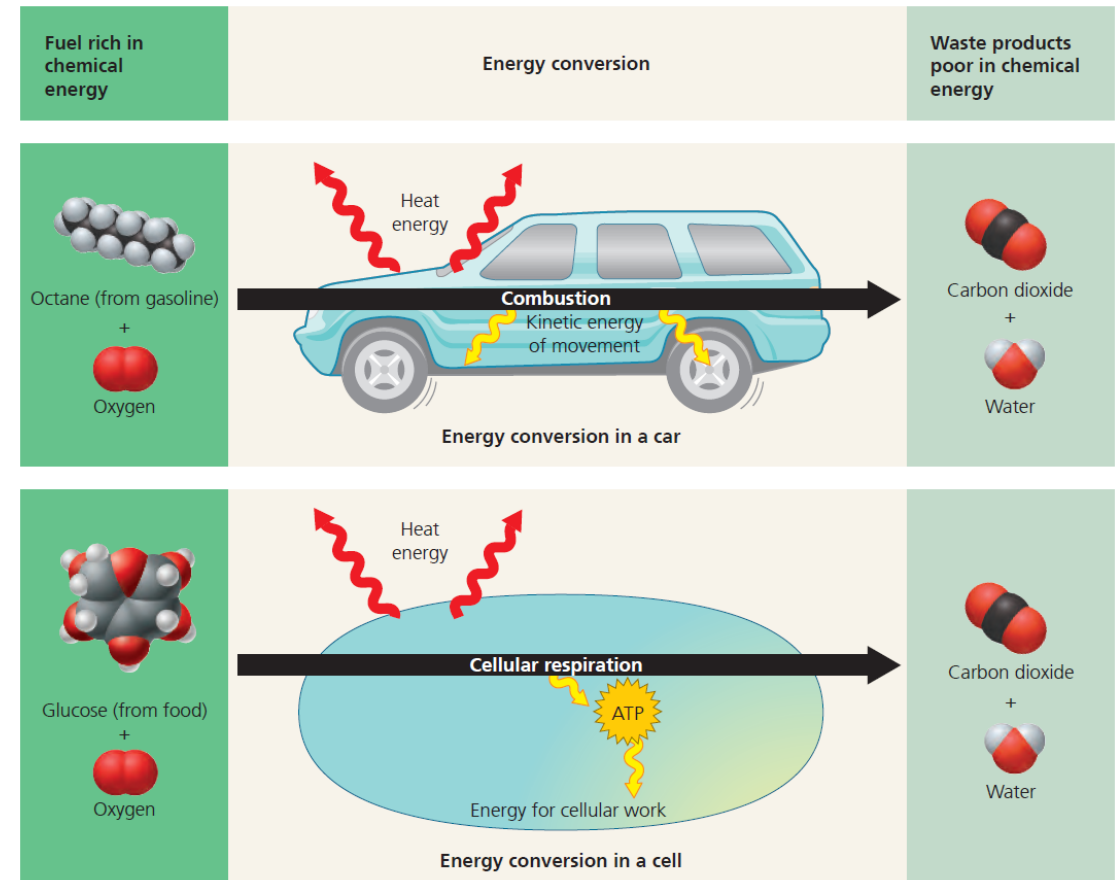


Learning Objectives 1

- Explain why photosynthesis is important to most ecosystems.
- Define and compare autotrophs and heterotrophs, producers and consumers.
- Explain how the processes of photosynthesis and cellular respiration are complementary to each other.

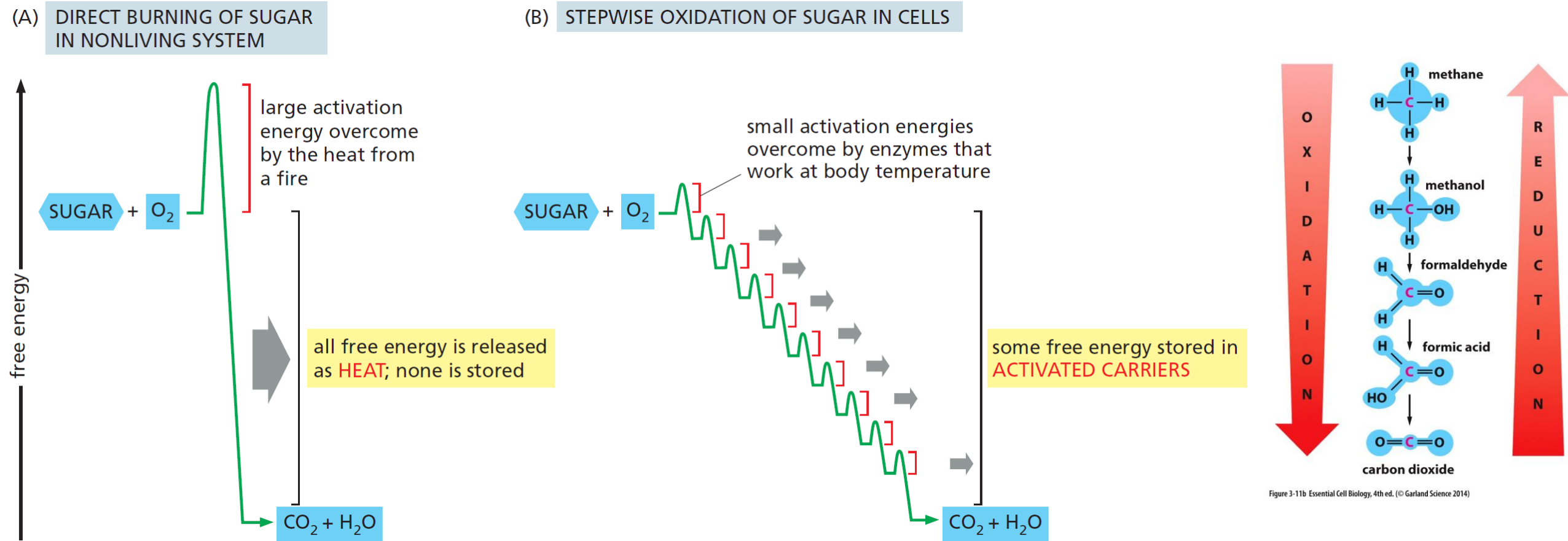
Chemical Energy

- The molecules of food, gasoline, and other fuels have a form of **potential energy** called chemical energy, which arises from the **arrangement of atoms** and can be released by a chemical reaction.
- In both cases, this process breaks organic fuel into **smaller waste molecules** that have much less chemical energy than the fuel molecules did, thereby releasing energy that can be used to perform work.



Stepwise Oxidation of Sugar vs. Burning

- Humans convert about **34% (1/3)** of our food energy to useful work.
- The **rest** of the energy released generates **body heat**.

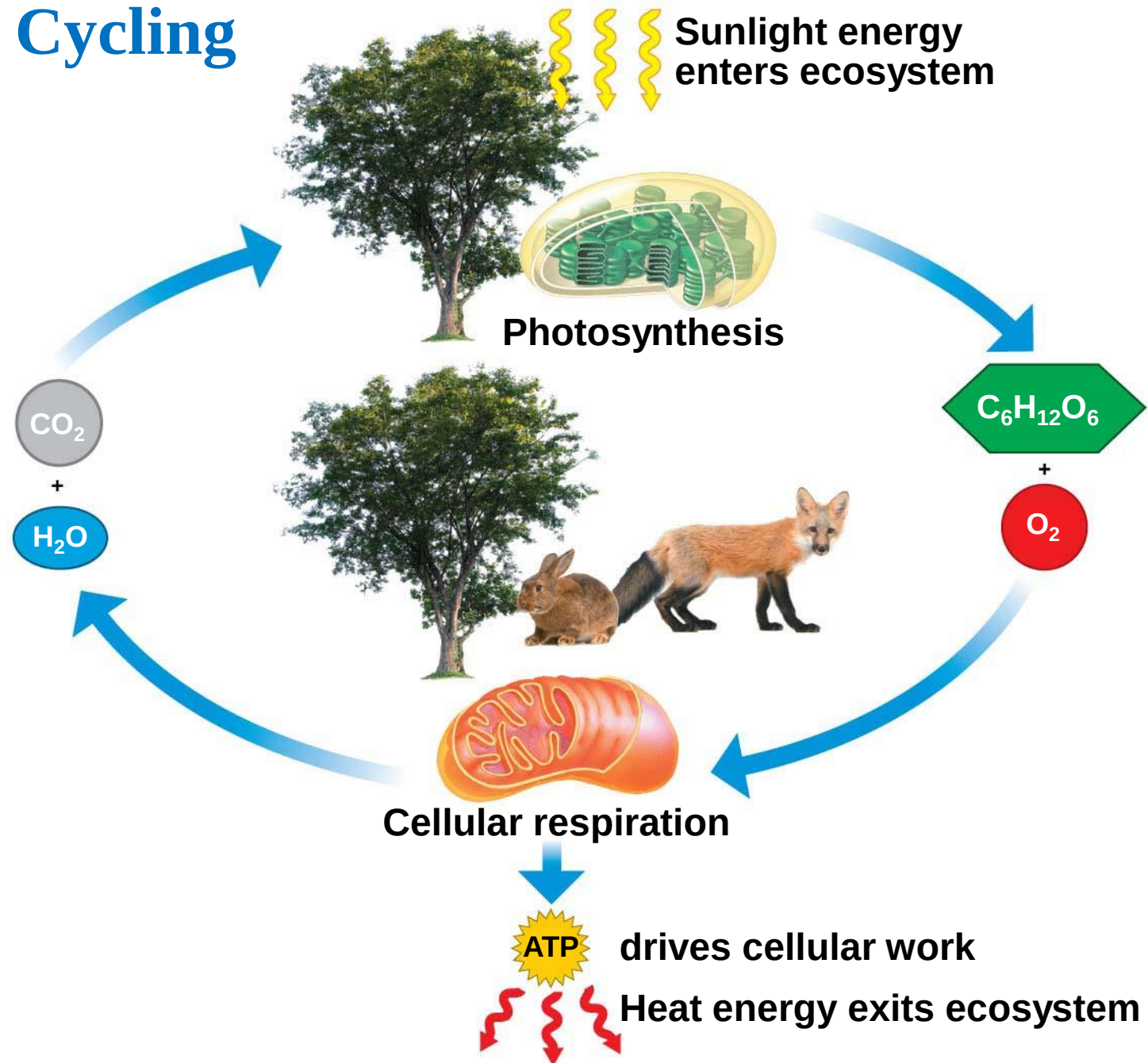


Producers and Consumers

- All life requires energy!
- Plants and other **auto**trophs (自养生物) (“**self**-feeders”) are organisms that make all their own organic matter, including carbohydrates, lipids, proteins, and nucleic acids, from nutrients that are entirely inorganic:
 - carbon dioxide from the air and water
 - minerals from the soil
- **Hetero**trophs (异养生物) (**other**-feeders)
 - include humans and other animals
 - cannot make organic molecules from inorganic ones.

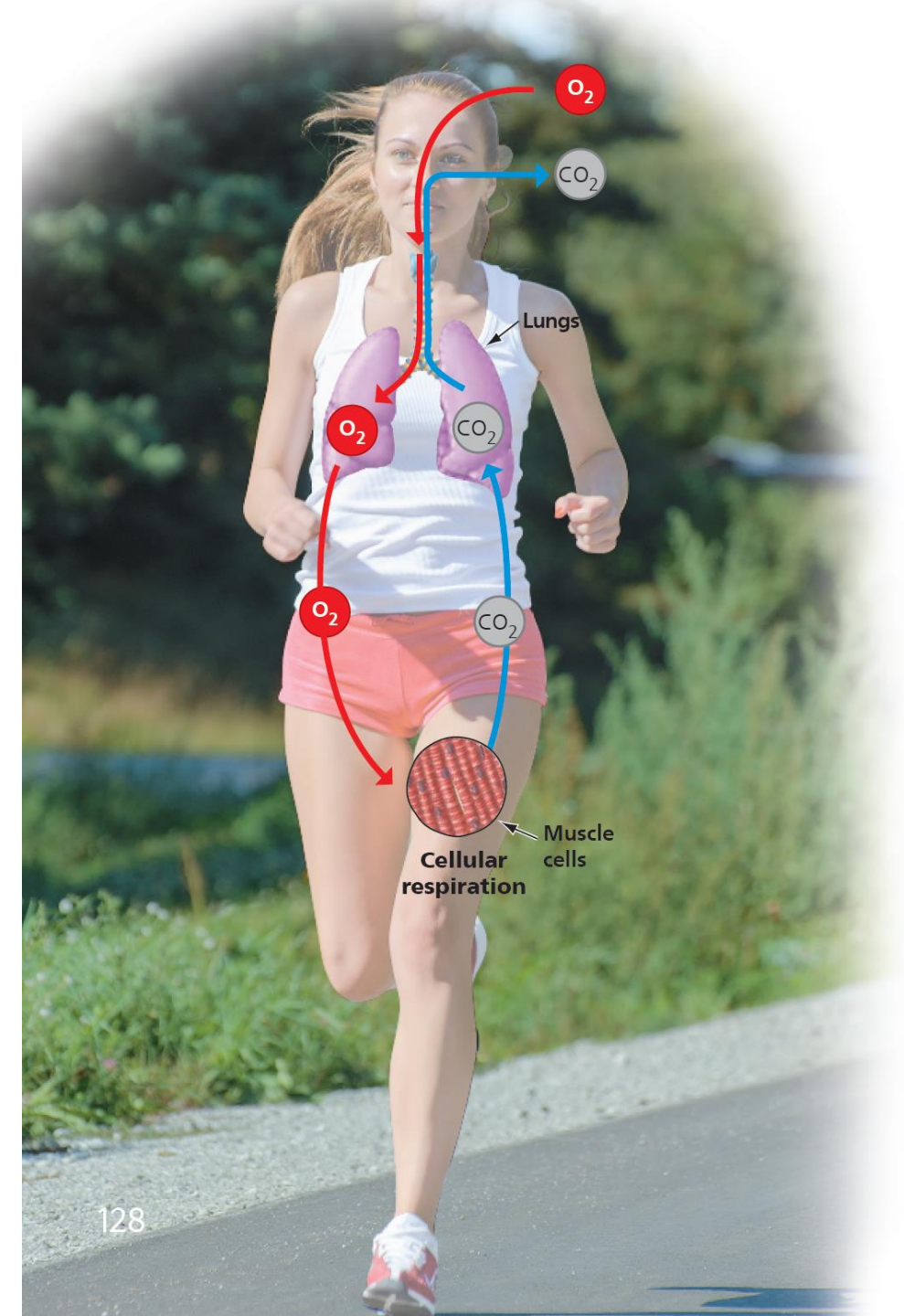
Energy Flow and Chemical Cycling

- Photosynthesis and cellular respiration form chemically complementary cycling.
- Plants usually make **more** organic molecules than they need for fuel.
 - provides material for the plant to grow or
 - can be stored, as starch in potatoes



Cellular Respiration and Breathing

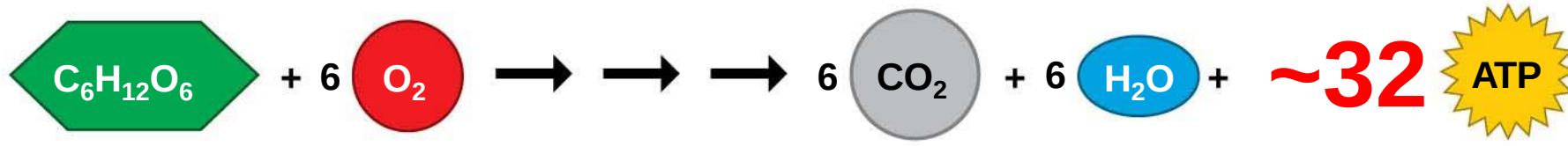
- Aerobic harvesting of chemical energy from organic fuel molecules
 - an aerobic process—it requires oxygen
- The main way that chemical energy is harvested from food and converted to ATP



Learning Objectives 2

- Write and explain the overall equation for cellular respiration.
- Explain how electrons are transferred from glucose to NADH to an electron transport chain to oxygen.
- Compare the reactants, products, location in the cell, and energy yield of the three stages of cellular respiration (glycolysis, citric acid cycle, and electron transport chain).

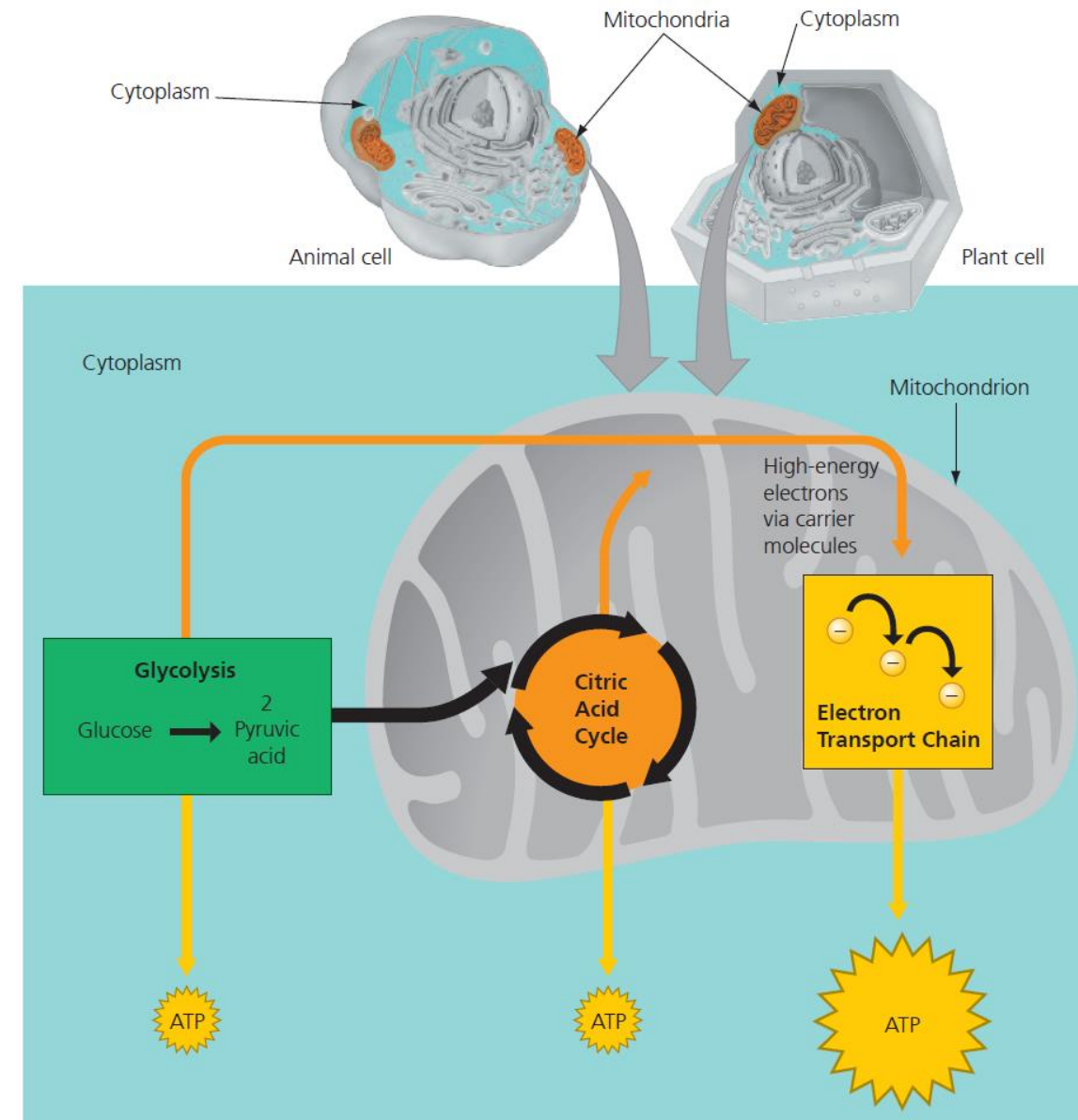
An Overview of Cellular Respiration



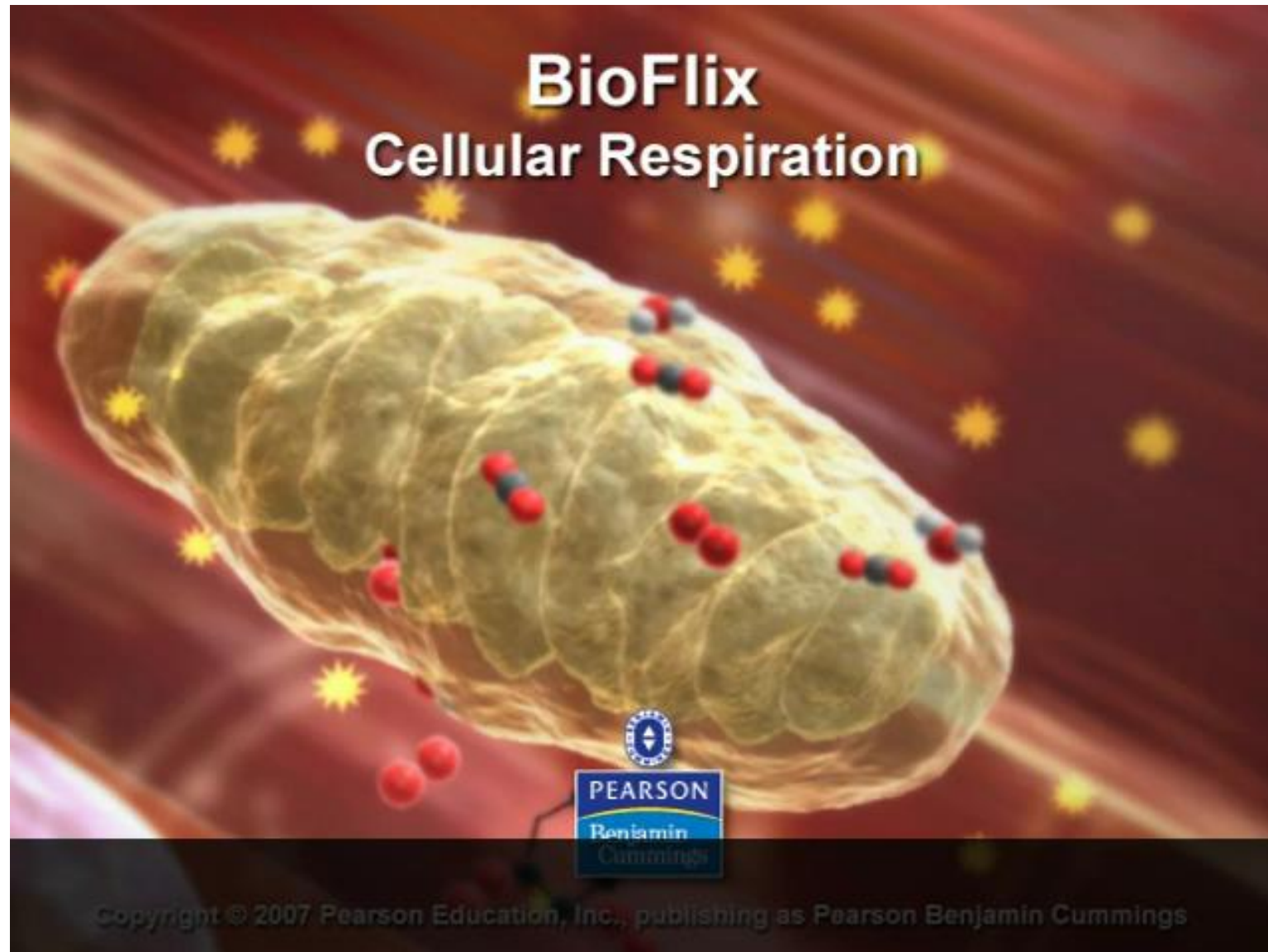
- Consists of many chemical steps, with more than two dozen reactions in all.
- Involves a specific **enzyme** to catalyze each reaction.
- Constitutes **one of the most important metabolic pathways** for nearly every eukaryotic cell.
- Provides the energy these cells need to maintain the functions of life.

Three Main Stages of Cellular Respiration

- Glycolysis (糖酵解),
- Citric acid cycle (CAC)
 - Tricarboxylic acid (TCA) cycle
 - Krebs cycle
- Electron transport

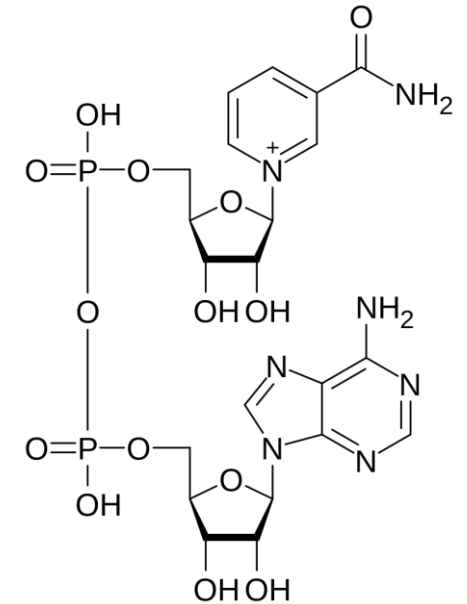
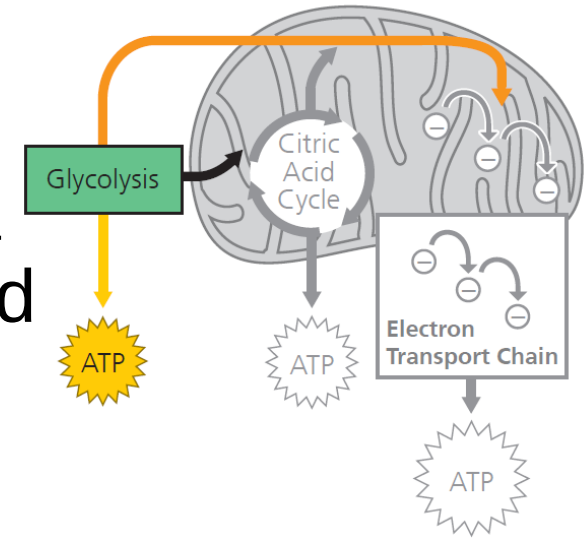


Cellular Respiration

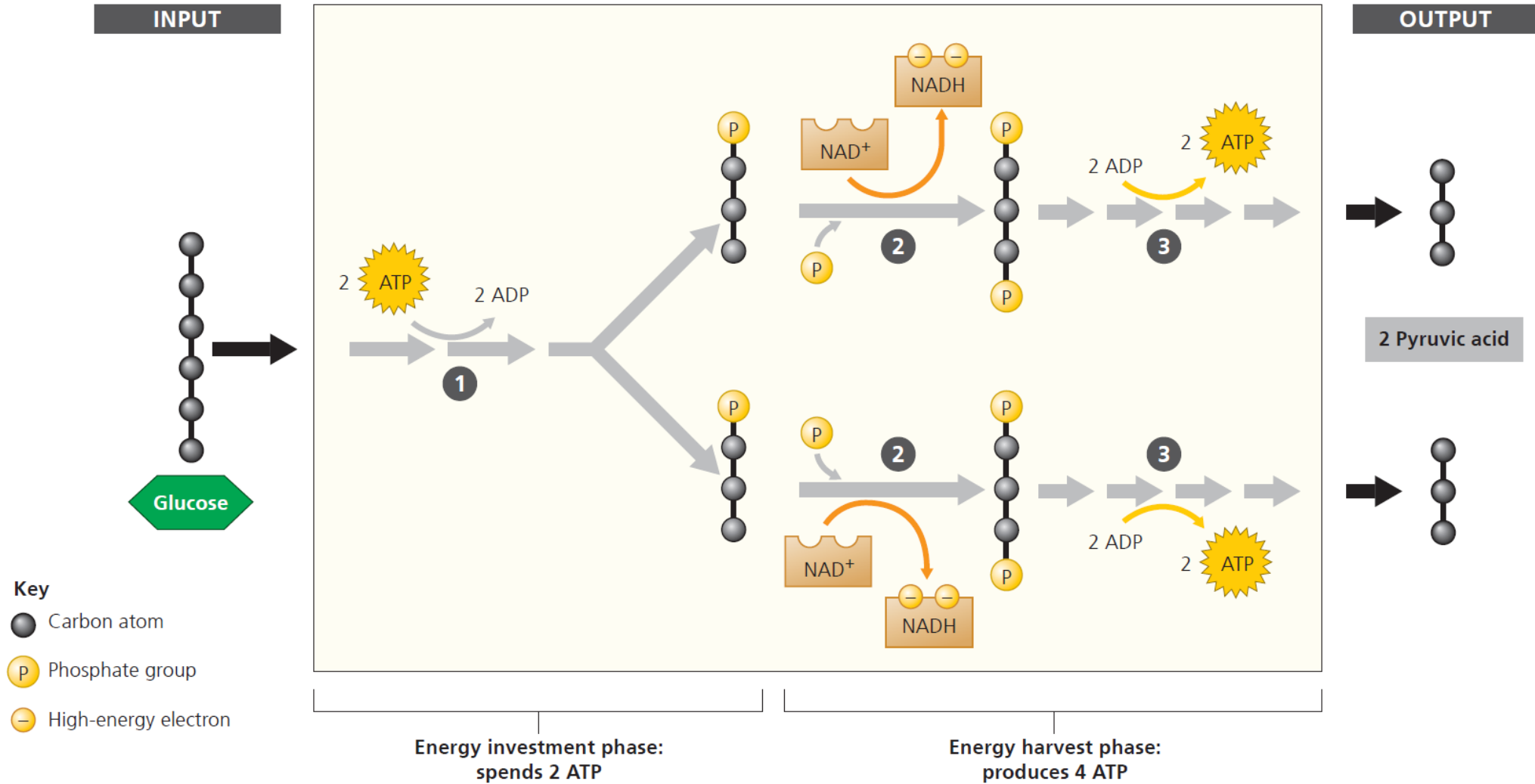


Stage 1: Glycolysis

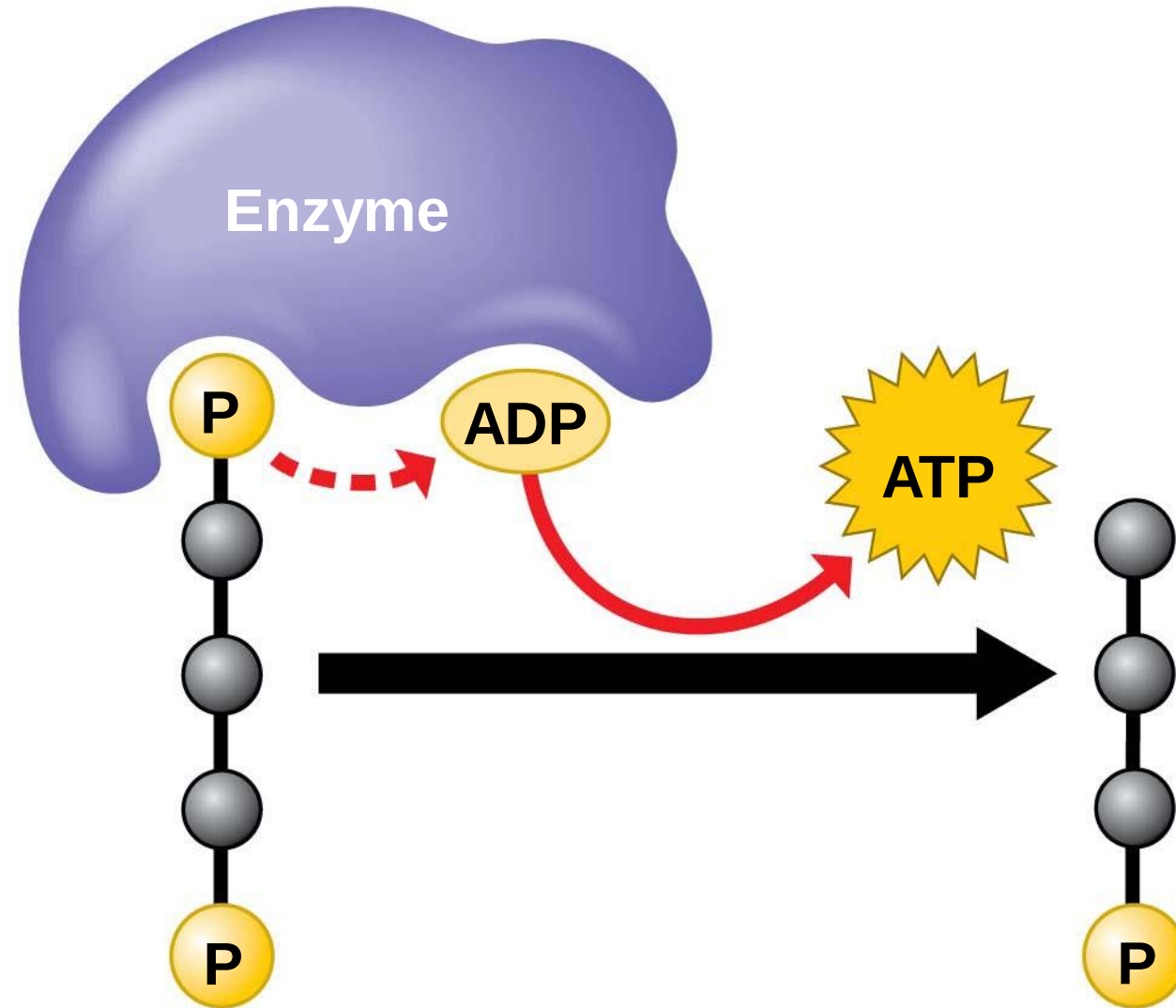
- A molecule of glucose is split into **two molecules** of a compound called pyruvic acid (丙酮酸), usually located **in the cytoplasm**.
- Generate a small amount of ATP **directly**
- And much more ATP **indirectly**, via reactions that transfer electrons from fuel molecules to a molecule called NAD⁺ (nicotinamide adenine dinucleotide) (烟酰胺腺嘌呤二核苷酸).
 - $\text{NAD}^+ + 2\text{e}^- + \text{H}^+ \rightarrow \text{NADH}$
 - NADH, as a shuttle carrying high-energy electrons from one area of the cell to another.



Stage 1: Glycolysis, 2 ATP and 2 NADH

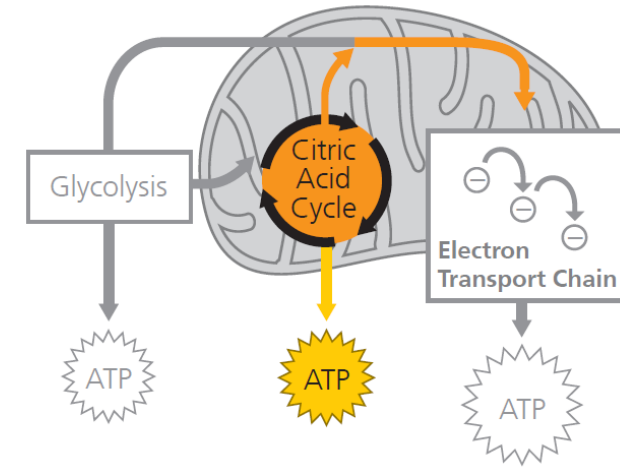


ATP Synthesis by Direct Phosphate Transfer: Substrate Level Phosphorylation



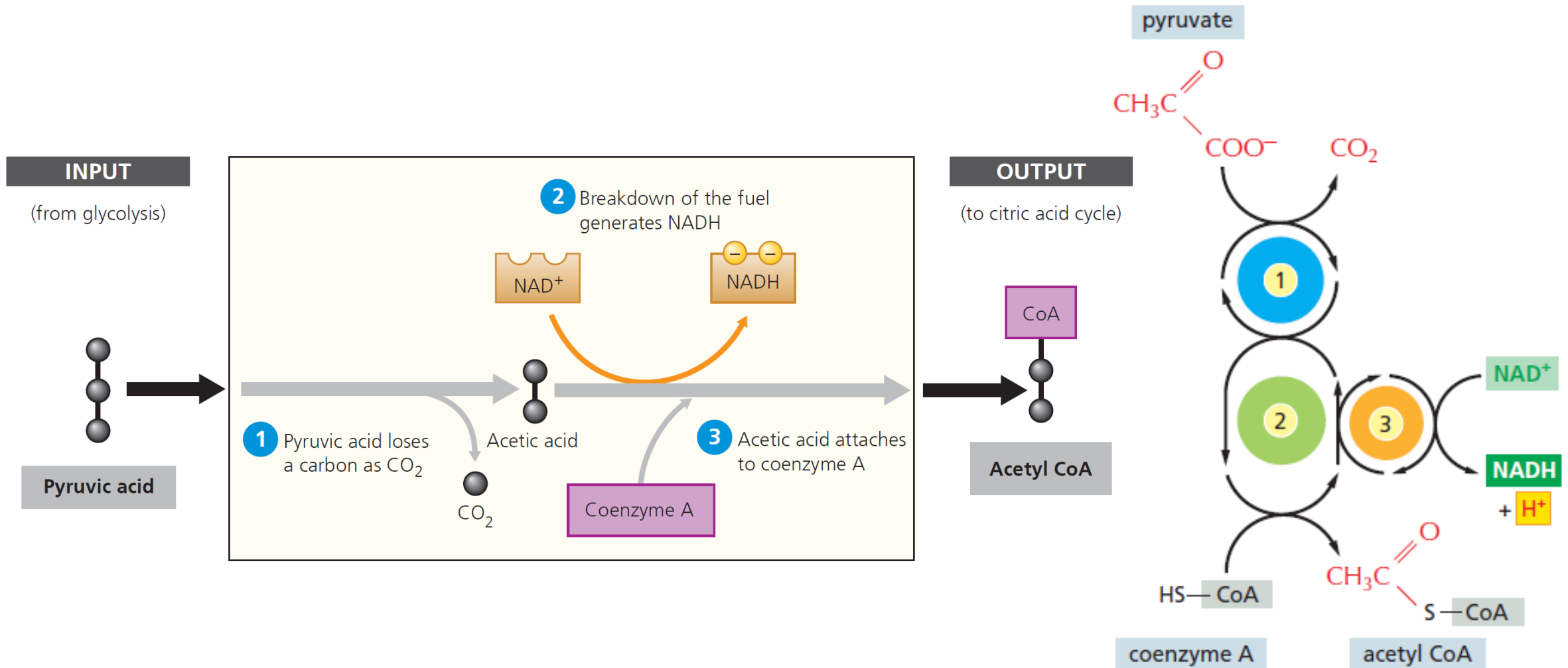
The Link Between Glycolysis and The Citric Acid Cycle

- Each pyruvic acid loses **a carbon** as CO_2 . The remaining fuel molecules, with only two carbons left, are called acetic acid (乙酸).
- Electrons are stripped from these molecules and transferred to another molecule of NAD^+ , forming more NADH.
- Finally, each acetic acid is attached to a molecule called coenzyme A (CoA) (辅酶A) to form acetyl CoA (乙酰辅酶A).
- The CoA escorts the acetic acid into the first reaction of the citric acid cycle and is then stripped and **recycled**.



The Link Between Glycolysis and The Citric Acid Cycle

2 CO₂ and 2 NADH



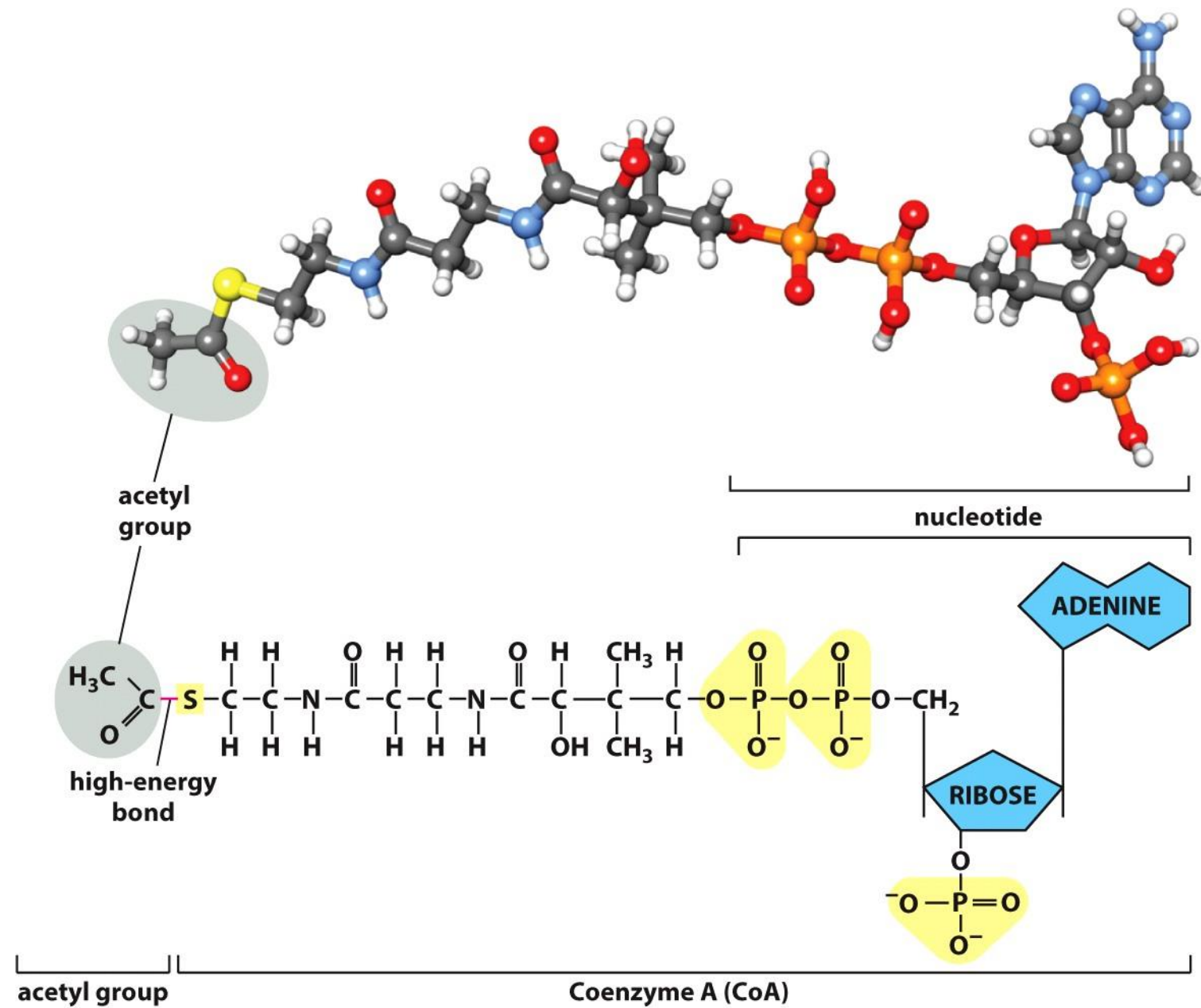
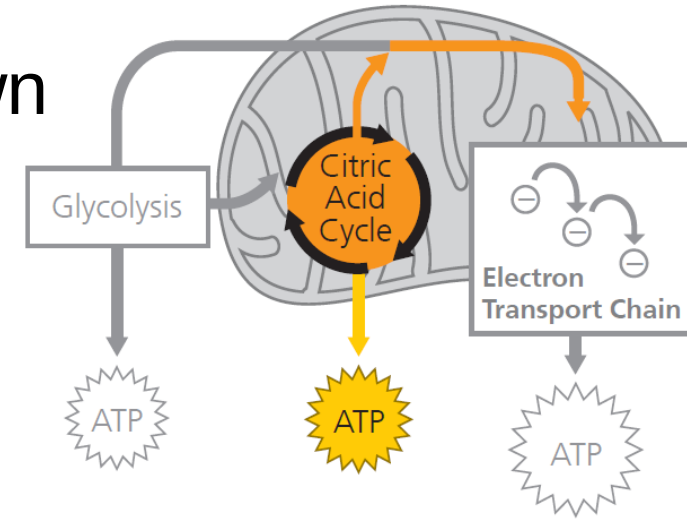


Figure 3-36 Essential Cell Biology, 4th ed. (© Garland Science 2014)

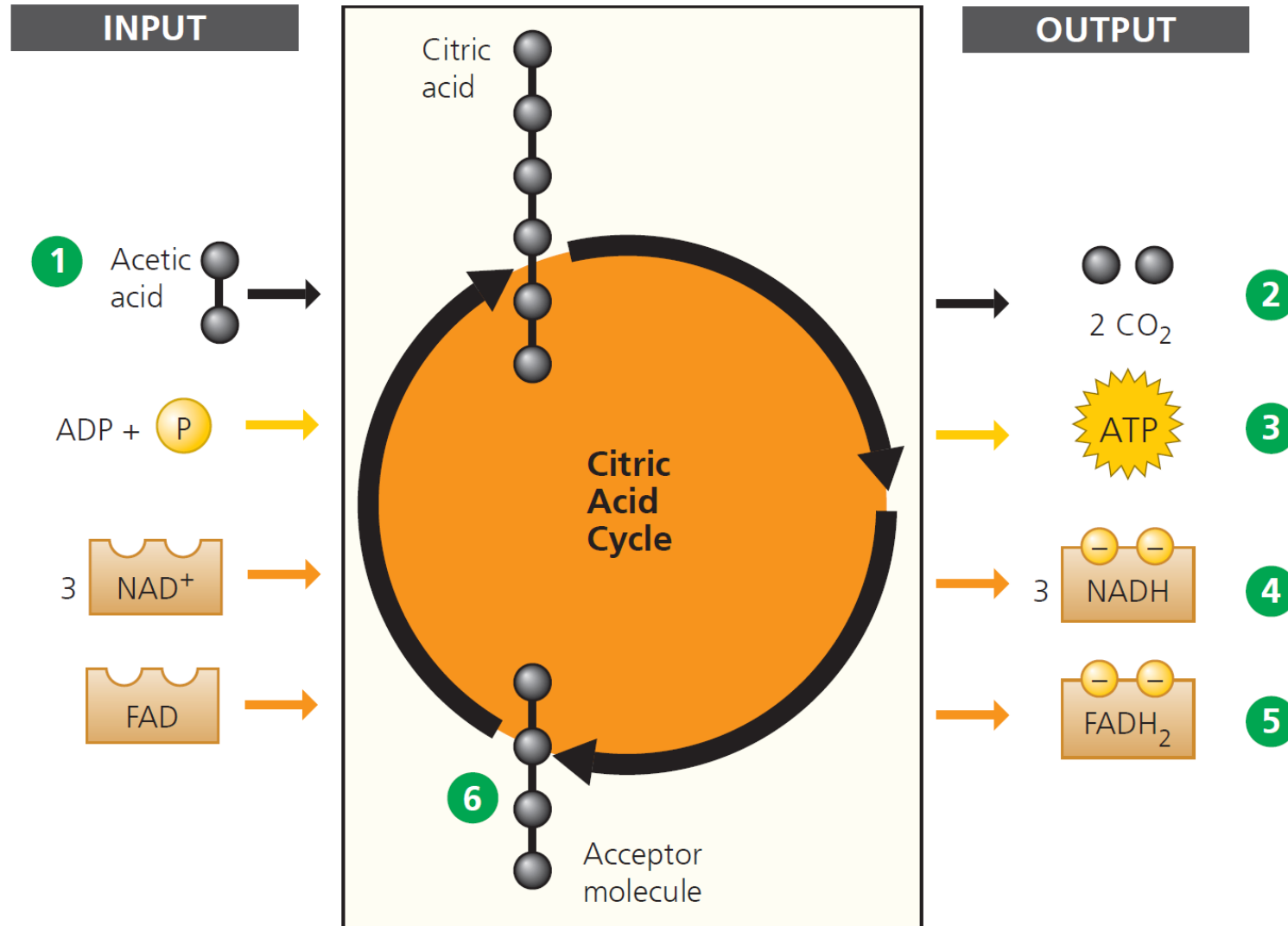
Stage 2: The Citric Acid Cycle

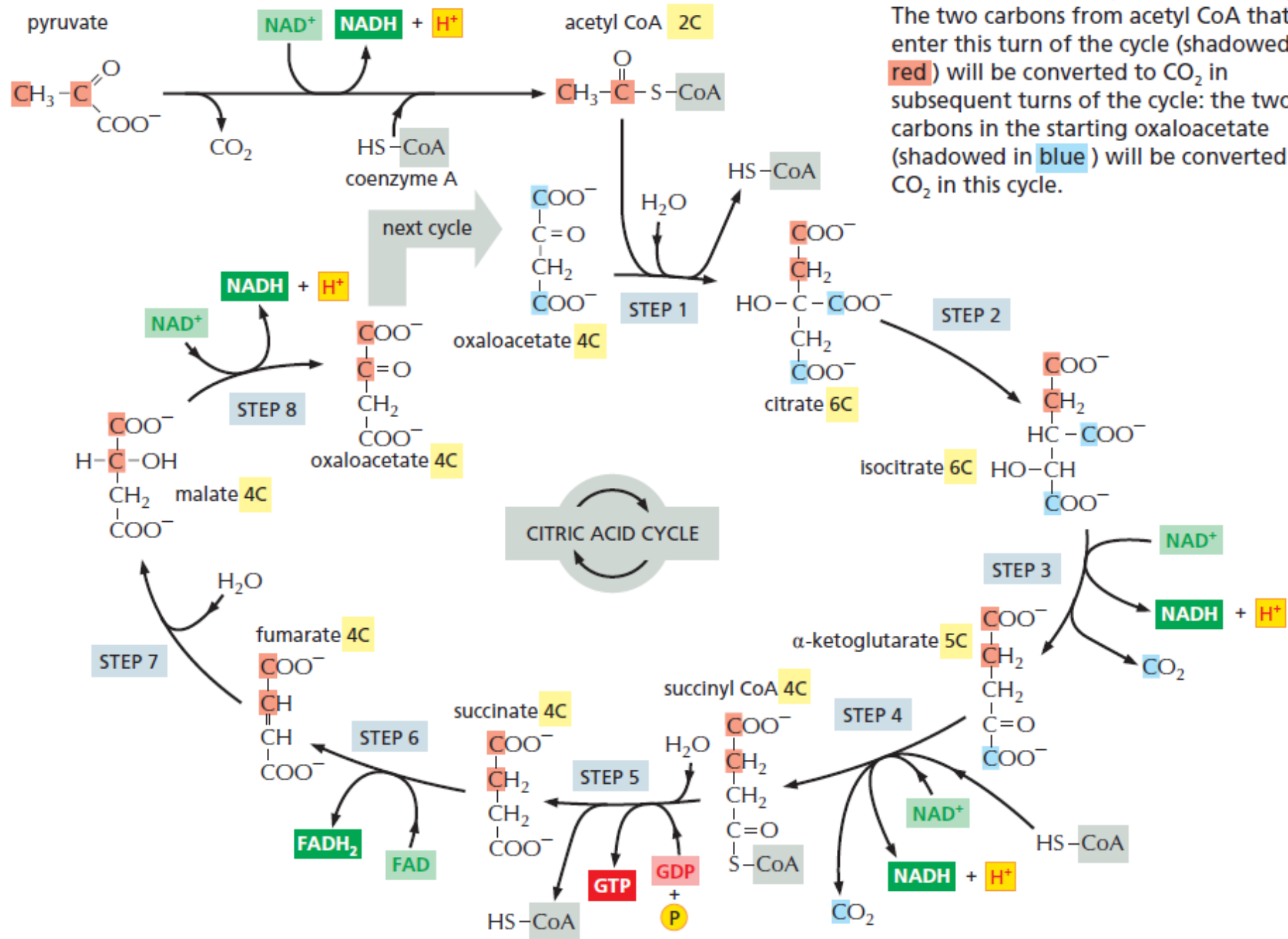
- Dismantle the acetic acid molecules all the way down to CO_2 , eventually exhaled.
- Acetic acid joins a four-carbon acceptor molecule to form a **six-carbon product**, **citric acid** (for which the cycle is named).
- For every acetic acid molecule that enters the cycle as fuel, two CO_2 molecules **eventually exit** as a waste product.
- Some of the energy is used to produce ATP directly.
- Much more energy in the form of NADH and a closely related electron carrier called FADH_2 .



Stage 2: The Citric Acid Cycle

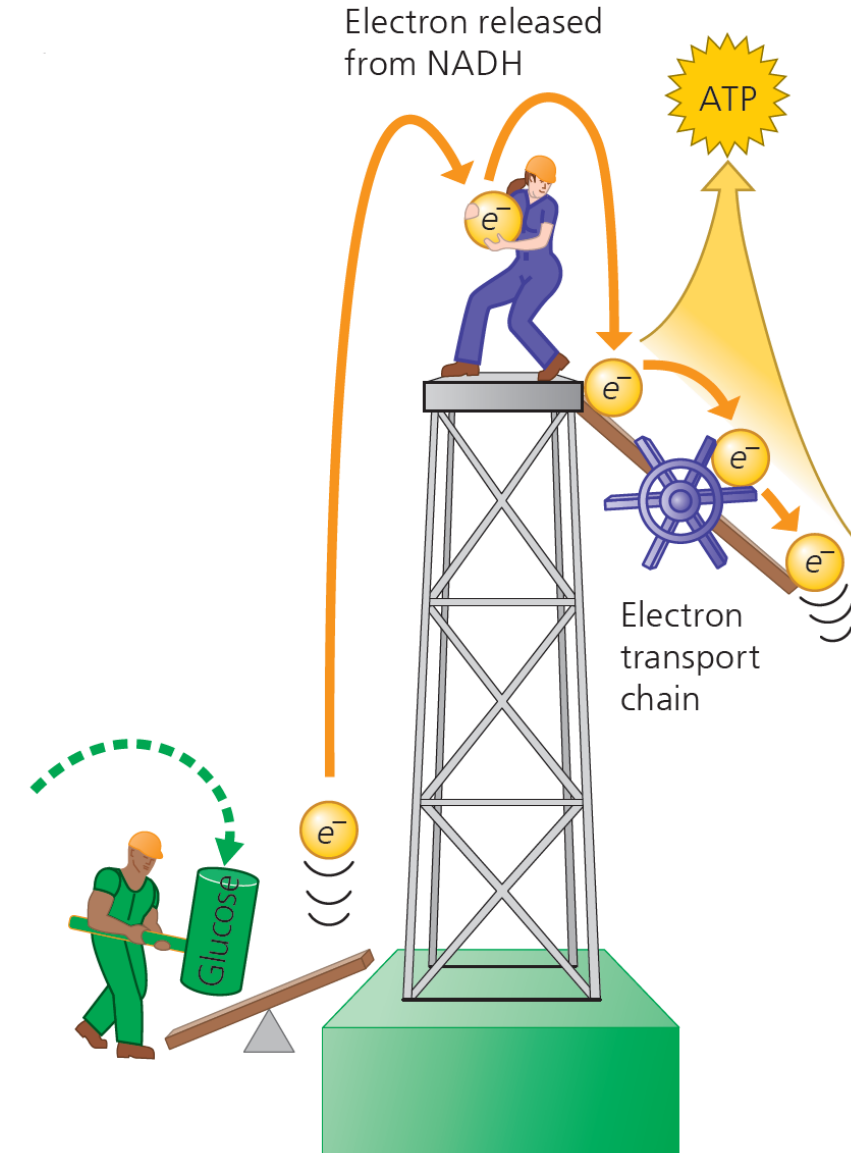
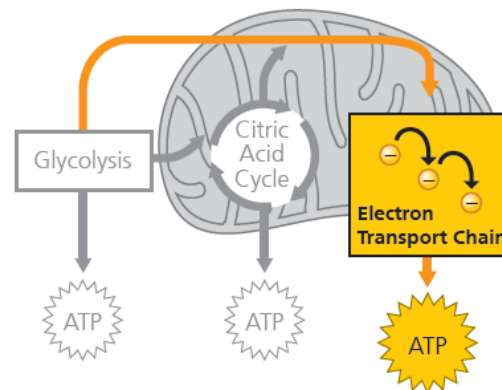
4 CO₂, 2 ATP, 6 NADH, and 2 FADH₂

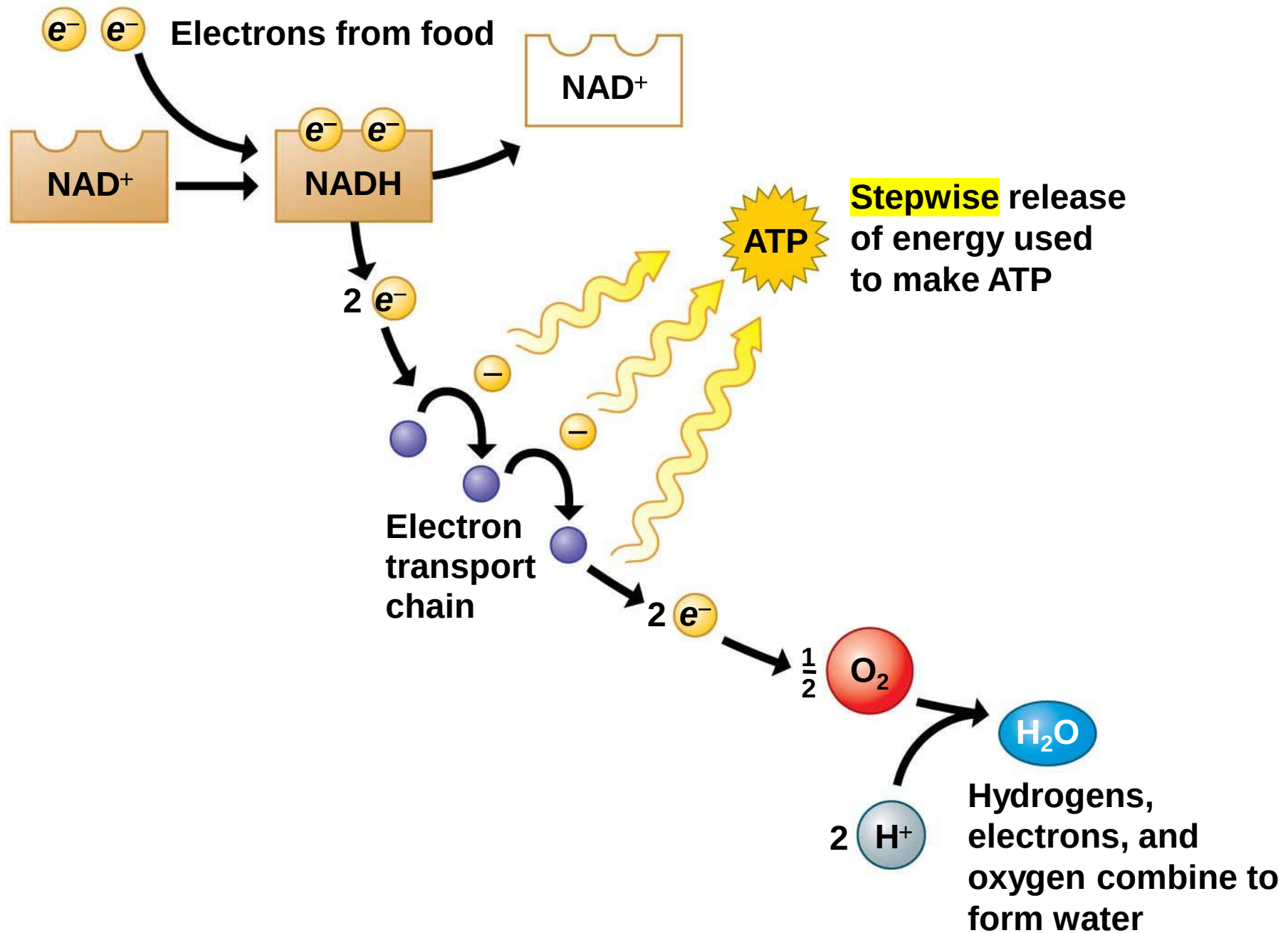




Stage 3: Electron Transport

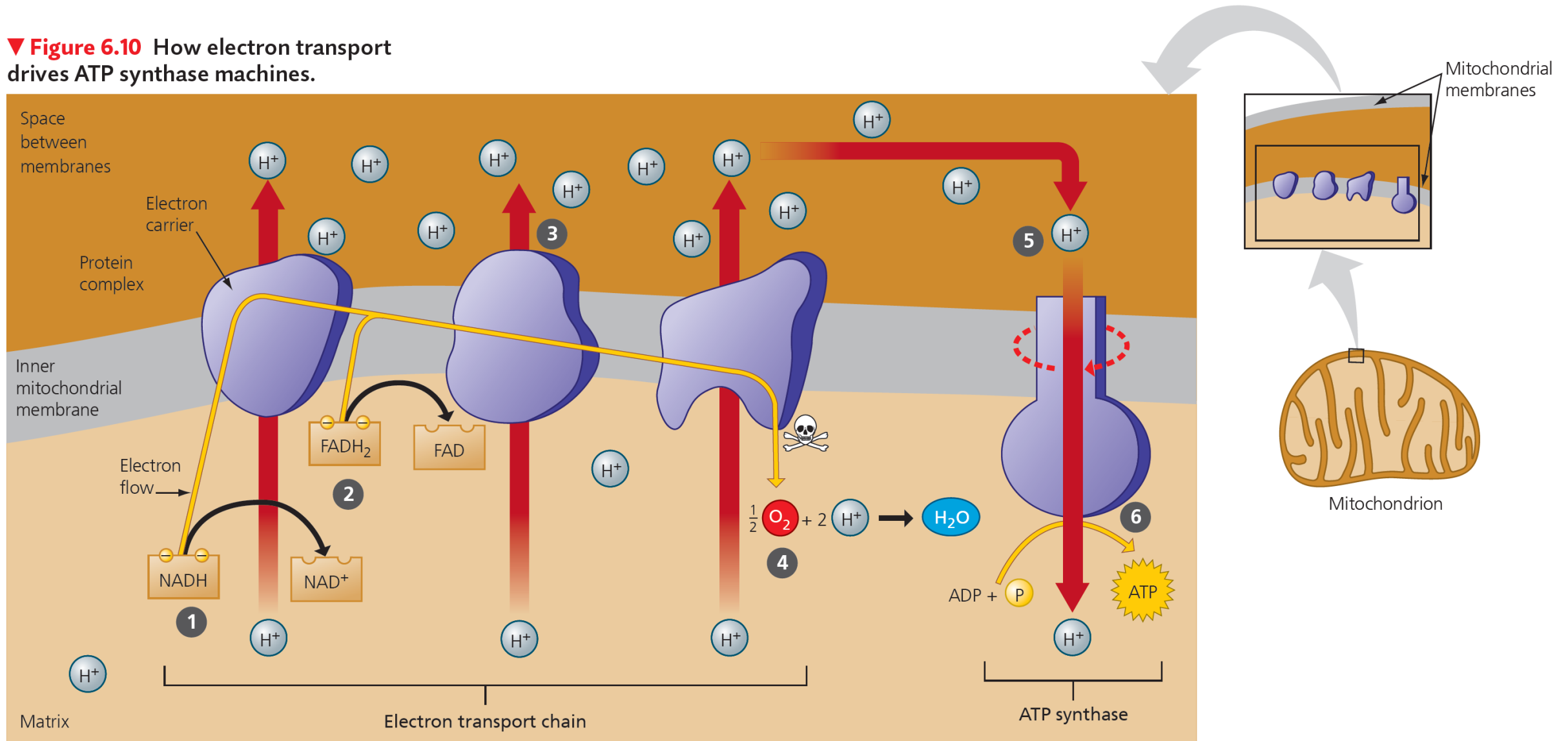
- Electrons captured from food by NADH are stripped of their energy, **a little bit at a time**, until they are finally **combined with oxygen to form water**.
- The proteins and other molecules that make up electron transport chains are embedded within the **inner membrane of the mitochondria**.





How electron transport drives ATP synthase machines

▼ **Figure 6.10** How electron transport drives ATP synthase machines.



Electron-Transport Chain

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Electrochemical Gradient

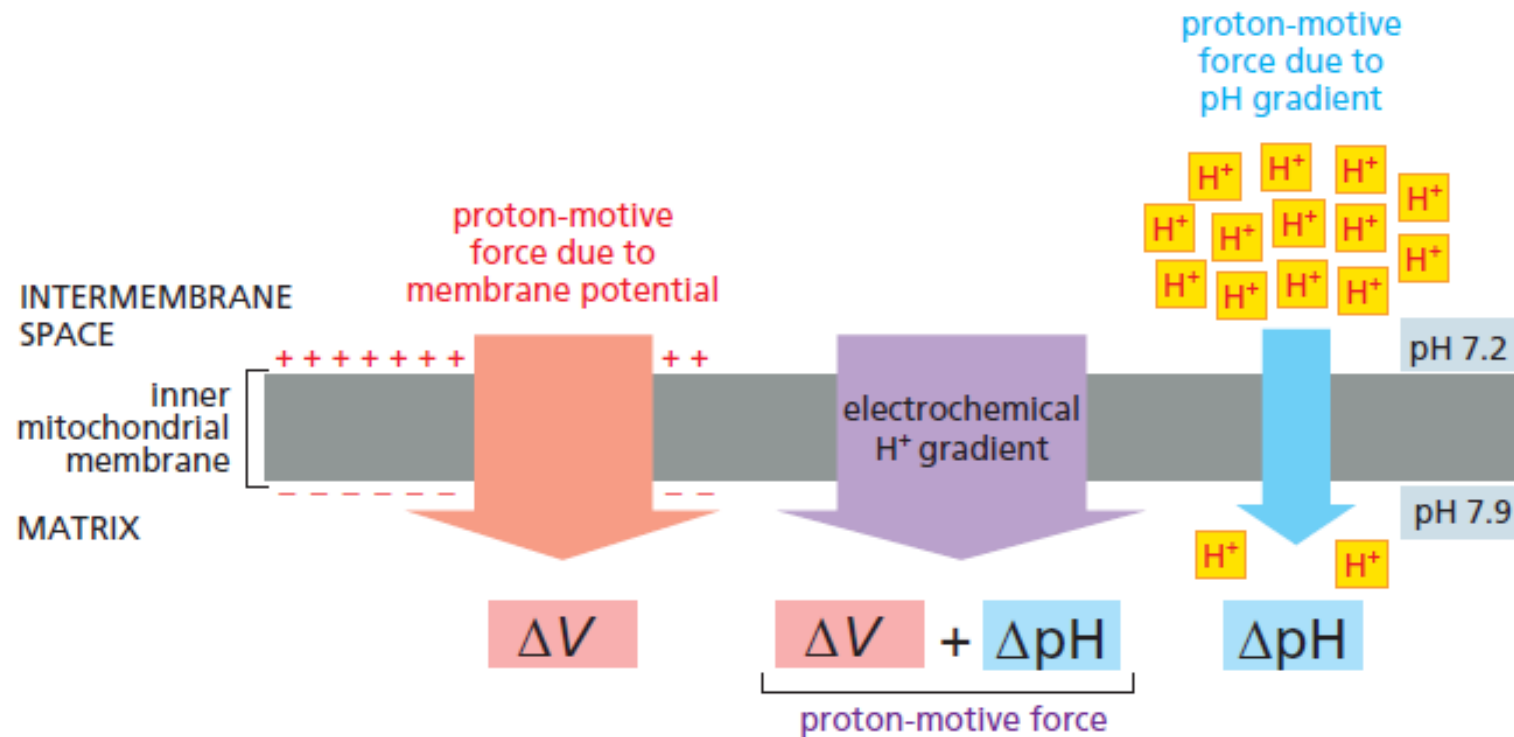
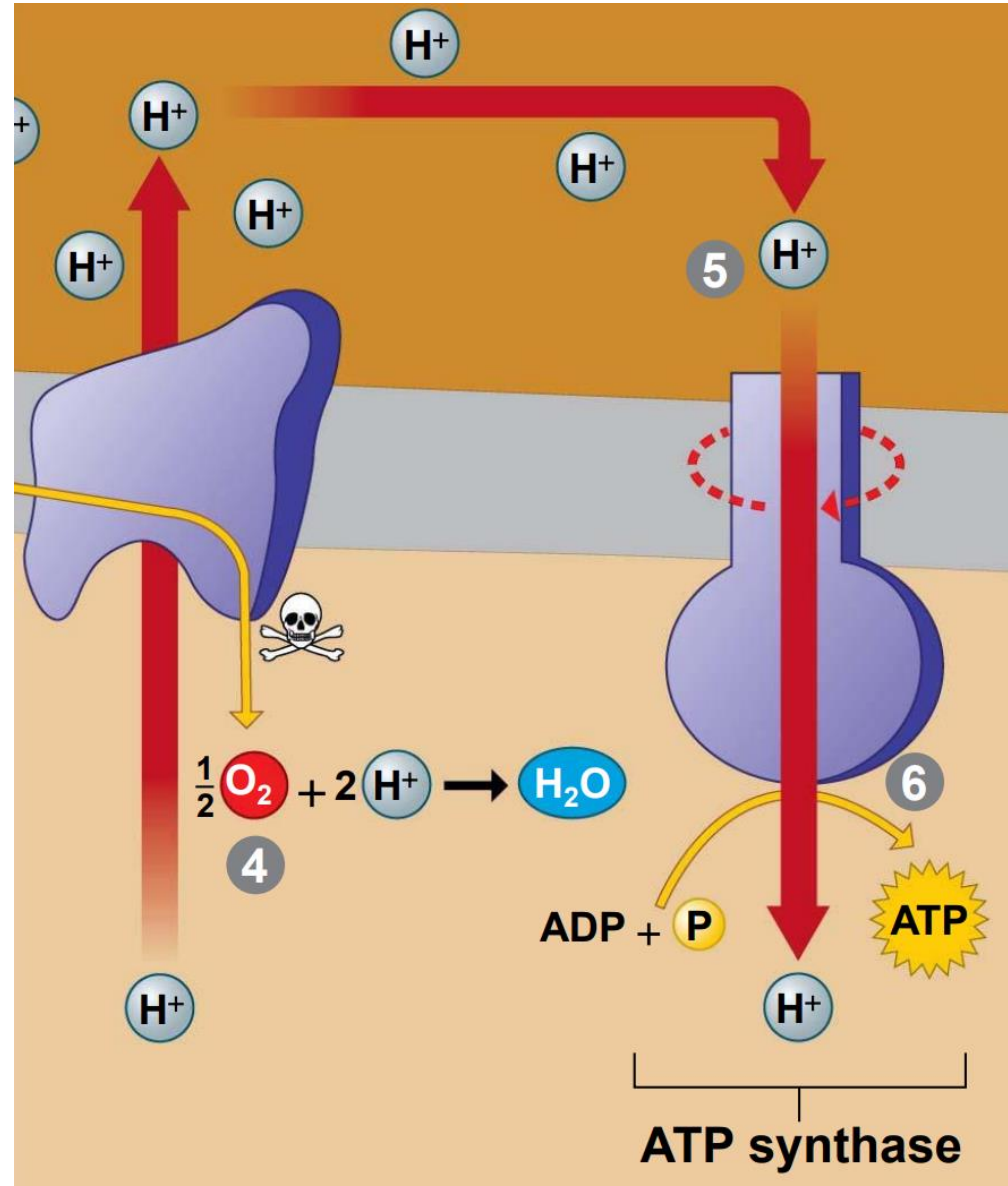


Figure 14–15 The electrochemical H^+ gradient across the inner mitochondrial membrane includes a large force due to the membrane potential (ΔV) and a smaller force due to the H^+ concentration gradient—that is, the pH gradient (ΔpH). The intermembrane space is slightly more acidic than the matrix, because the higher the concentration of protons, the more acidic the solution (see Panel 2–2, pp. 68–69). Both the membrane potential and the pH gradient combine to generate the proton-motive force, which pulls H^+ back into the mitochondrial matrix. The exact, mathematical relationship between these forces is expressed by the Nernst equation (see Figure 12–24).

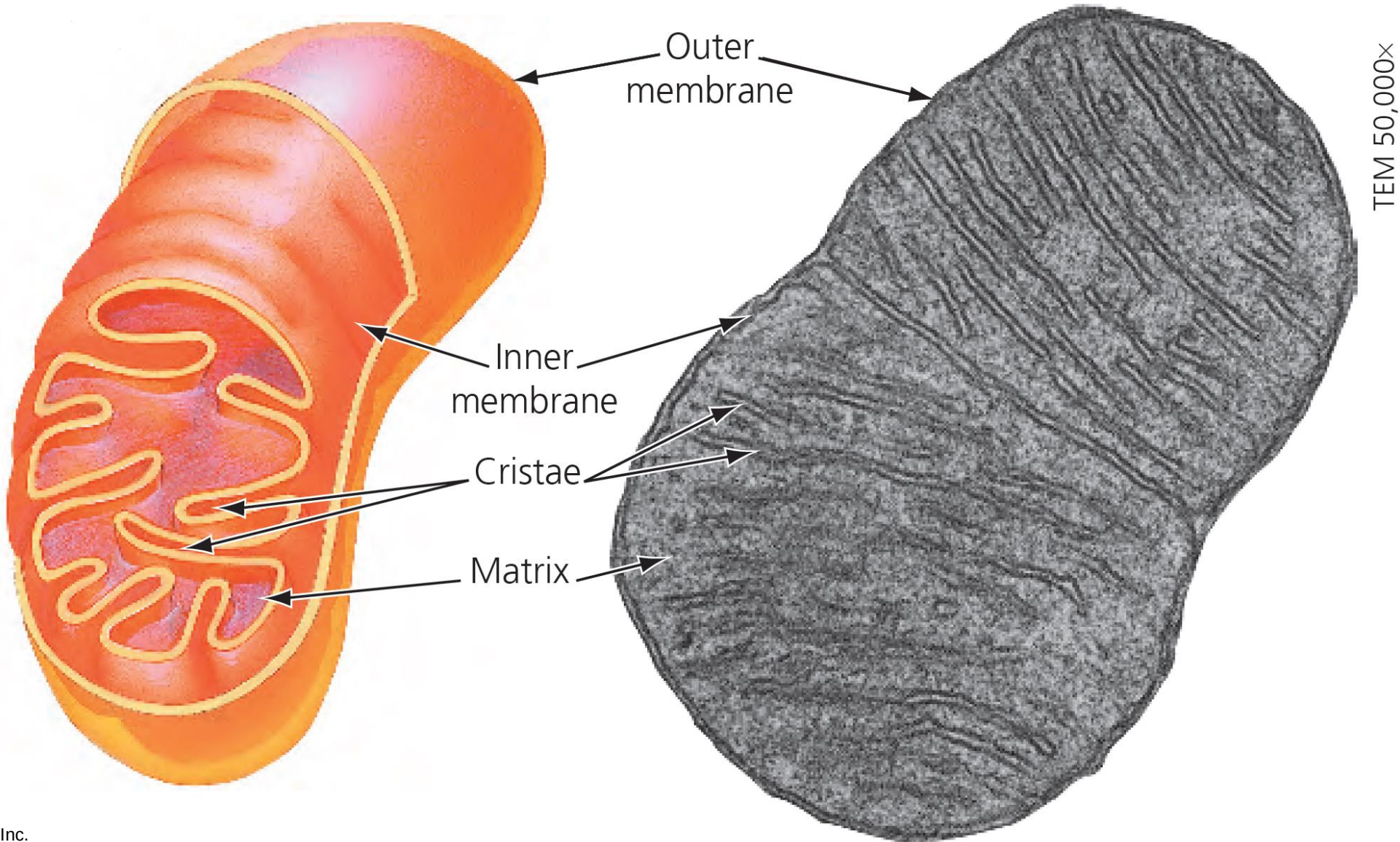
Oxidative Phosphorylation



ATP Synthase— A Molecular Turbine

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Function of the Highly Folded Inner Membrane



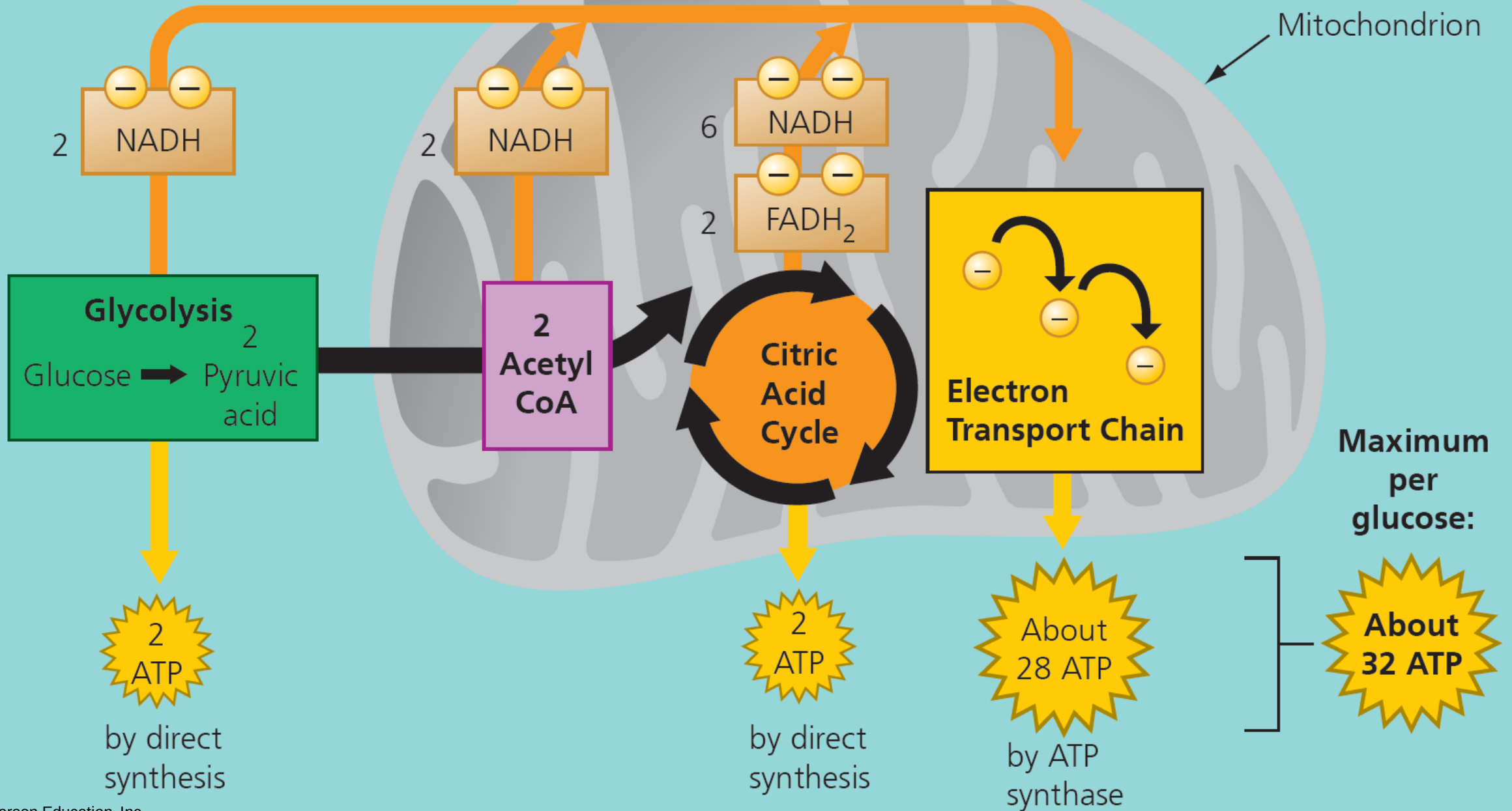
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ATP Synthase—Dimers in Cristae

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Cytoplasm

Mitochondrion



▼ **Figure 6.12 Energy from food.** The monomers from carbohydrates (polysaccharides and sugars), fats, and proteins can all serve as fuel for cellular respiration.

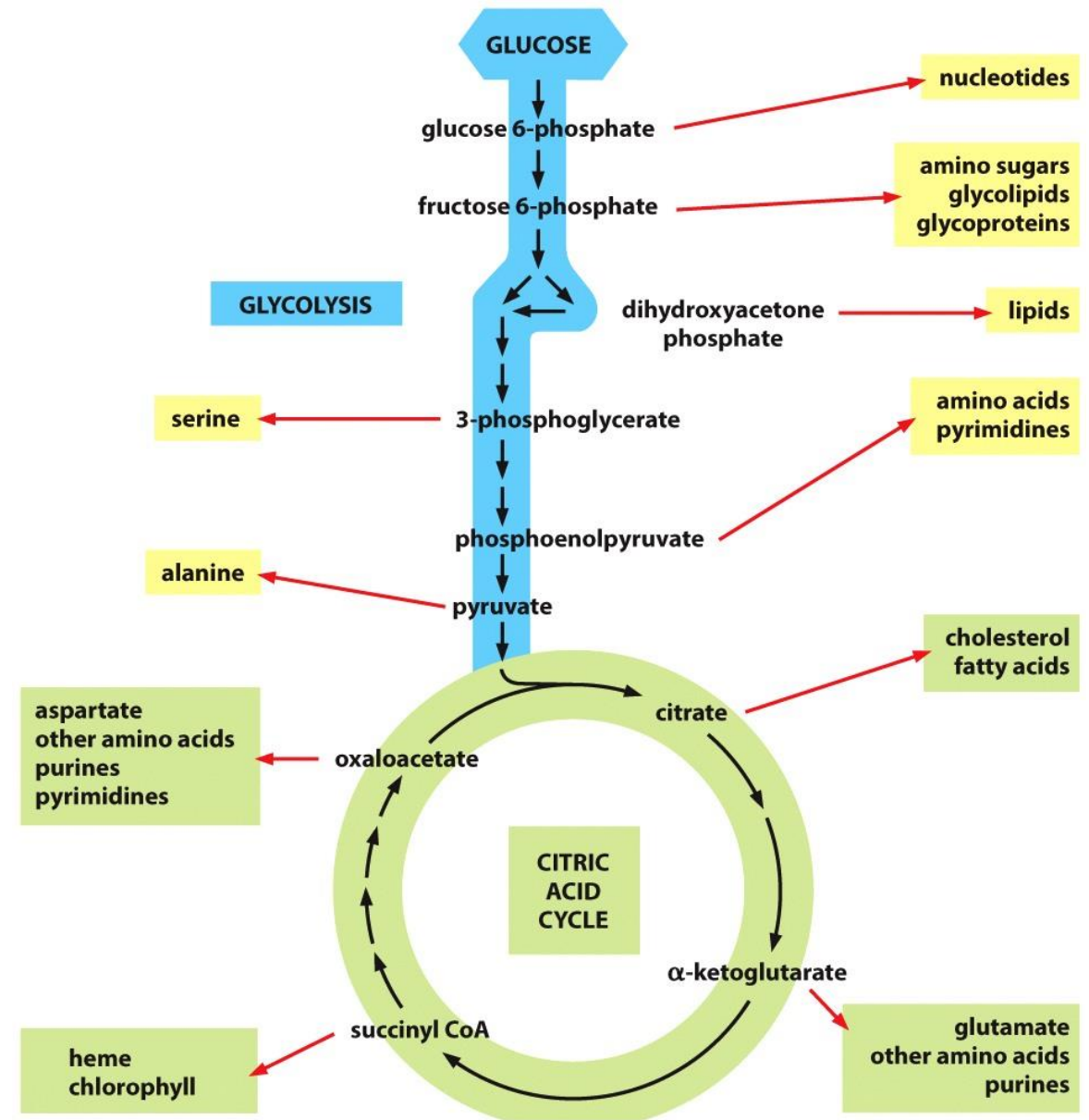
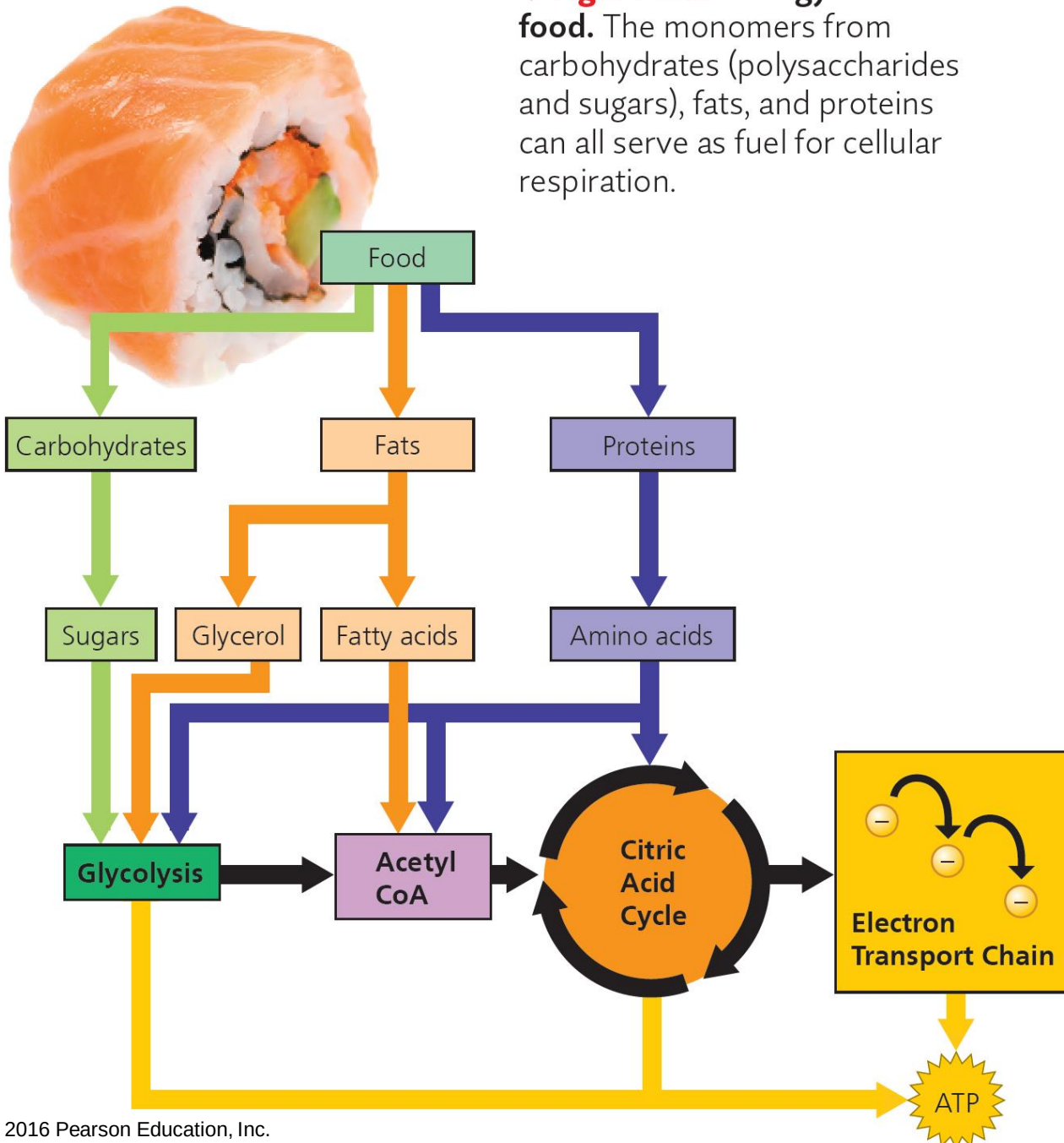


Figure 13-14 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Learning Objectives

- Compare the reactants, products, and energy yield of aerobic metabolism versus fermentation in human cells including where each occurs
- Compare fermentation in humans to that in microorganisms
- Explain how our current understanding of the effect of lactic acid buildup in muscles demonstrates the tentativeness of scientific conclusions and why conclusions are not considered “proof.”

Fermentation (发酵):

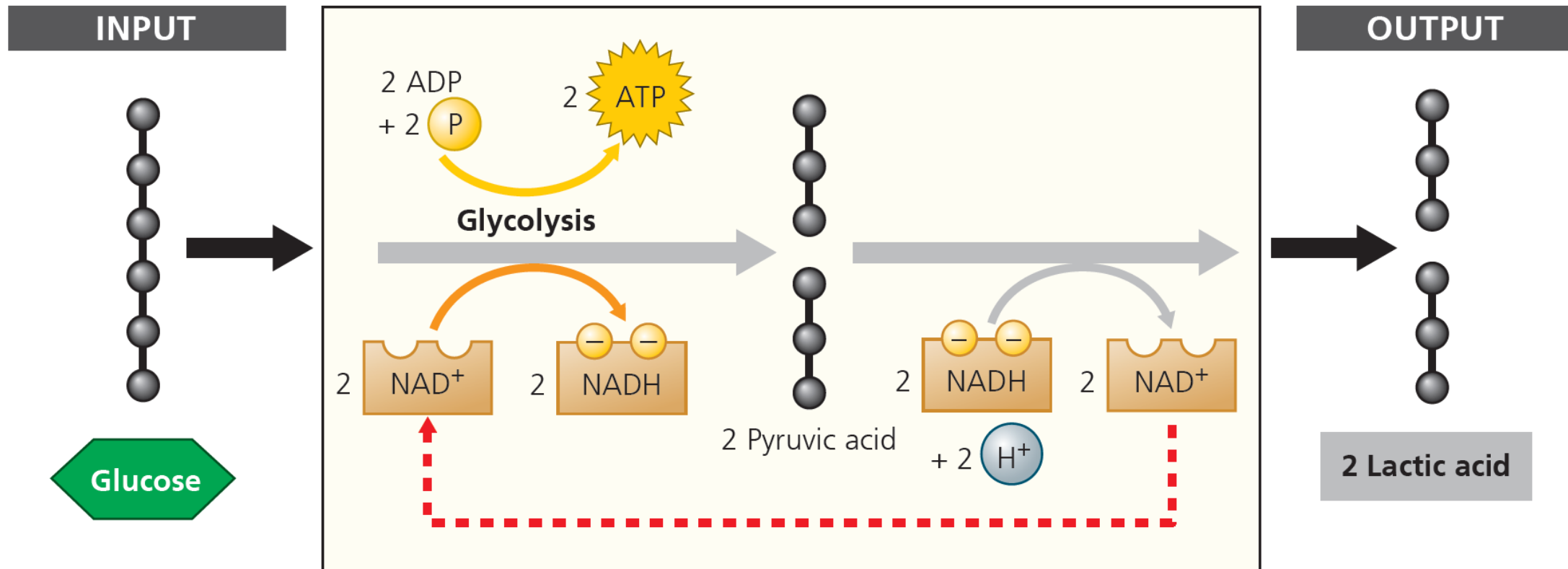
Anaerobic (without oxygen) Harvest of Food Energy

- Used by
 - Our muscle cells in times of temporary **oxygen shortage**.
 - Microorganism, employed by us to produce foods and beverages.
- Fermentation relies on **glycolysis** to produce ATP.
 - does not require oxygen
 - produces **two ATP** molecules for each glucose broken down to pyruvic acid.

Fermentation in Human Muscle Cells Produces Lactic Acid

After functioning anaerobically for about 15 seconds, muscle cells begin to generate ATP by the process of fermentation.

“emergency mode”

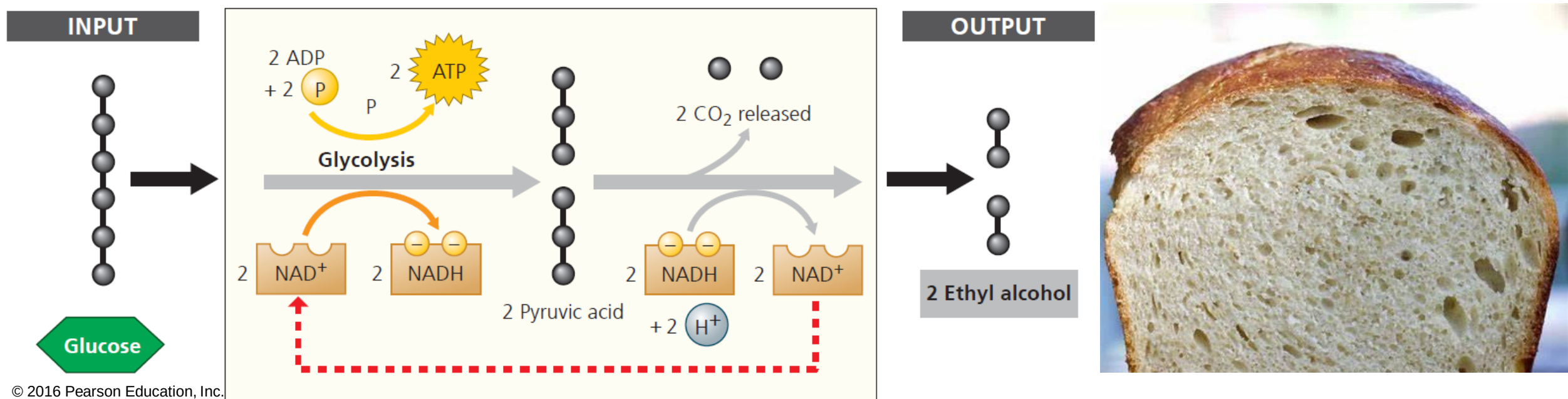


Fermentation in Microorganisms

- Fermentation alone is enough to sustain many microorganisms.
- Wastes from fermentation can be lactic acid, ethyl alcohol, or other compounds, depending on the species.
- The lactic acid produced by some microbes using lactic acid fermentation is used to produce
 - cheese, sour cream, and **yogurt**
 - soy sauce, **pickles**, cabbage, and olives
 - sausage meat products

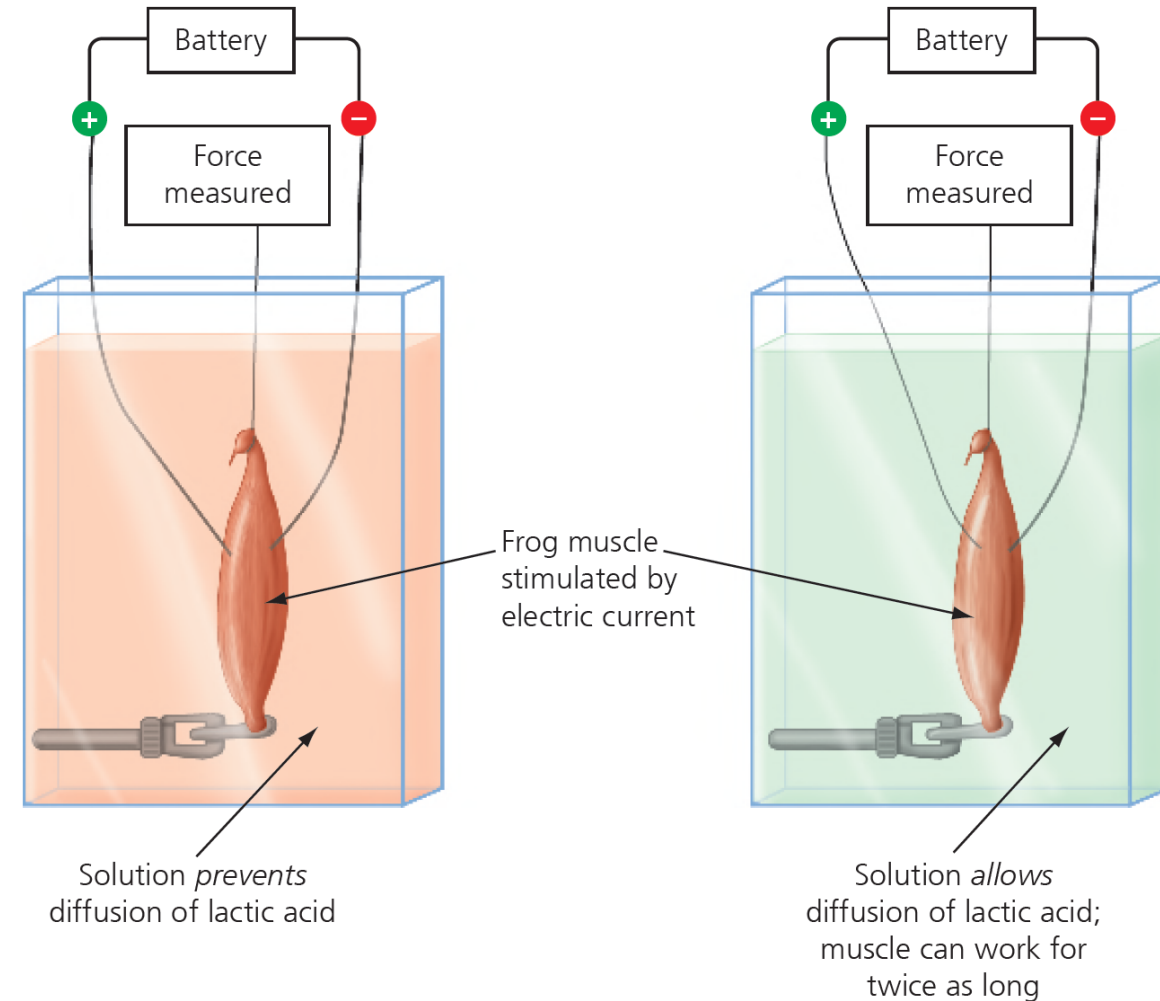
Fermentation in Yeast Produces Ethyl Alcohol

- Yeast is capable of cellular respiration and fermentation.
 - Perform **alcoholic fermentation** to produce CO_2 and ethyl alcohol.
- Yeast for alcoholic beverages production: beer and wine
- Rising bread dough by the CO_2 bubbles from fermenting yeast.



Muscle Burn in Exercising Science by A.V. Hill (1922 Nobel Prize for his investigations of muscle contraction)

- Hypothesis: The buildup of lactic acid would cause muscle activity to stop.
- Results: When lactic acid was allowed to diffuse away, performance improved significantly.
- Conclusion: The buildup of lactic acid is the primary cause of muscle failure under anaerobic conditions.



The Tentativeness of Scientific Conclusions

- Evidence began to accumulate that contradicted Hill's results.
 - The effect that Hill demonstrated did not appear to occur at human **body temperature**.
 - Certain individuals who are unable to accumulate lactic acid have muscles that fatigue more rapidly, the opposite of what is expected.
- Recent research indicates that increased levels of other ions may be to blame.
- Science is dynamic and subject to constant adjustment as new evidence is uncovered.